MAIS PAC

IPhO 02.05: Hydrodynamics and Surface Tension · Worked Solutions

Hydrodynamics and Surface Tension: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

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Solution 1 — Pressure in the deep. Return →

(a) p_({gauge}) = ρ gh = 1030×9.81×10^4 = 1.01×10^8 {Pa}, about 1000 atmospheres.

(b) F = pA with A = π(0.15)² = 0.0707 {m²}: F_({gauge}) = 7.1×10^6 {N}; including the atmosphere adds only 1.0×10^5×0.0707 ≈ 7×10³ {N}, a 0.1% correction.

(c) The glass sees the **difference** across it: the outside is at atmospheric pressure, so the load is essentially the gauge pressure. The total pressure would be relevant if the porthole were sealed and evacuated on one side and the whole assembly were at depth. **Check:** the gauge pressure dominates here by three orders of magnitude, which is why designers quote gauge values for deep-sea structures.

Solution 2 — Floating and sinking. Return →

(a) Floating means displaced weight equals the block's weight: ρ_fV_({sub})g = ρ_bVg, so V_({sub})/V = ρ_b/ρ_f = 600/1000 = 0.60.

(b) In ethanol the ratio is 600/789 = 0.76, so three-quarters of the block is submerged.

(c) The ratio ρ_b/ρ_f contains no volume, so shape and size are irrelevant. In a liquid of exactly equal density the block floats fully submerged and neutral — any further increase in ρ_f makes it sink. **Check:** the fractions must lie between 0 and 1, which they do for ρ_b<ρ_f.

Solution 3 — Torricelli and the jet. Return →

(a) Bernoulli between the free surface (speed ≈0) and the hole, both at the same pressure p_0:

$$p_0 + \rho\, g(h+H) = p_0 + (1)/(2)\rho\, v^2 + \rho\, gH \quad v = \sqrt{(2gh)}. \tag{1}$$

The approximation is A≫ a, so the free surface descends negligibly fast.

(b) The jet leaves horizontally at height H: t = √(2H/g) and R = v t = 2√(hH).

(c) With h + H = fixed, R = 2√(h(-h)), maximal at h = /2, i.e. the hole halfway down. For a comparable with A the surface speed cannot be neglected: the exact result is v = √(2gh),/√(1-(a/A)²). **Check:** R is symmetric under h H, as the formula demands, and vanishes at both ends.

Solution 4 — The Venturi meter. Return →

(a) Continuity gives v_2 = v_1A_1/A_2; Bernoulli gives Δ p = (1)/(2)ρ(v_2² - v_1²). Substituting,

$$v_1 = \sqrt{((2\Delta\ p)/(\rho[(A_1/A_2)^2\text{-}1]))}, \quad Q = A_1 v_1. \tag{2}$$

(b) For A_2→ A_1 the denominator vanishes, so Q→0 for fixed Δ p — correct, since no pressure difference is generated. For A_2→0 the prediction diverges, signalling that real compressibility and viscosity dominate.

(c) (A_1/A_2)² = 6.25, so v_1 = √(2(12000)/(1.2×5.25)) = √(3809) = 61.7 {m,s^(-1)} and Q = 0.020×61.7 = 1.23 {m³,s^(-1)}. **Check:** the units are m³/s and the speed is subsonic, so the incompressible treatment is borderline but defensible.

Solution 5 — Capillary rise. Return →

(a) h = 2γ/(ρ g r) r = 2γ/(ρ gh) = 2(0.072)/(1000·9.81·0.08) = 1.83×10^(-4) {m}, about 0.18 {mm}.

(b) Halving the radius doubles the rise to 16 {cm}.

(c) Mercury has a contact angle above 90°, so cos θ<0 and the formula predicts a **negative** height — a depression. The pressure just under the meniscus is p_0 - 2γ/r (a lower pressure for a concave meniscus), and that is a local quantity while the rise is the hydrostatic response to it. **Check:** the rise is inversely proportional to r, consistent with the pressure difference scaling as 1/r.

Solution 6 — The hydraulic press. Return →

(a) The areas are in the ratio $(20/2)^2 = 100$, so the force is $200 \times 100 = 2.0 \times 10^4$ {N}.

(b) Volume conservation: A_1d_1 = A_2d_2 d_1 = 100×0.05 = 5.0 {m}.

(c) Work in: 200×5.0 = 1000 {J}. Work out: 2.0×10^4×0.05 = 1000 {J}. Equal, so no energy is created: the press multiplies force and divides distance. **Check:** the mechanical advantage equals the area ratio, exactly as a lever's equals the arm ratio.

Solution 7 — The floating ice cube. Return →

(a) The ice displaces water whose weight equals the ice's weight, i.e. mass m. When it melts it becomes mass m of water, occupying exactly the volume it had displaced. The level is unchanged.

(b) An air bubble reduces the ice's mass without reducing its volume, so it displaces less water and the level **falls** slightly on melting. A dense stone inside makes the ice displace extra water equal to the stone's weight; once released, the stone sinks and displaces only its own volume, which is less than its weight's worth of water — so the level falls too. In both cases the level falls; only pure ice leaves it unchanged.

(c) Track displaced volume against weight before and after; water is the reference, and any material denser than water “wastes” displacement once it sinks. **Check:** the pure-ice case is the unique case where displaced mass and melted mass are equal.

Solution 8 — Bernoulli in a siphon. Return →

(a) Bernoulli between the free surface and the outlet, both at p_0: the height drop from the surface to the outlet is h_2 - h_1 (with the sign convention that the outlet is lower), giving $v = \sqrt{2g(h_2-h_1)}$.

(b) At the highest point, height H above the free surface, Bernoulli gives $p = p_0 - \rho gH - (1)/(2)\rho v^2$. For the water to stay liquid, p>0, hence $H\ p_0/(\rho g) \approx 10.3$ {m} (less with the kinetic term, and much less in practice because of dissolved gas).

(c) A bubble breaks the continuous liquid path: the pressure at the top cannot be transmitted through the gas in the same way, and the column separates. **Check:** setting v=0 in (b) returns the hydrostatic limit $p_0 = \rho gH$, the barometer height.

Solution 9 — Stokes' law and terminal velocity. Return →

(a) At terminal speed the forces balance: $6\pi r v_T = (4)/(3)\pi r^3(\rho_s-\rho_f)g$, so $v_T = (2r^2(\rho_s-\rho_f)g)/(9)$.

(b) $v_T \propto r^2$, so a bead twice as large falls four times faster.

(c) $v_T = (2(10^{-3})^2(2650-1000)(9.81))/(9(10^{-3}))) = 3.6\times10^{-3}$ {m,s^(-1)}. Then {Re} = ρ v d/ = $(1000)(3.6\times10^{-3})(2\times10^{-3})/10^{-3} = 7.2$, well below the turbulent transition and comfortably inside the Stokes regime. **Check:** the square-law scaling and the small Reynolds number together confirm that fine sediment settles slowly and smoothly.

Solution 10 — A barge, a bridge and a load. Return →

(a) $V_{sub} = M/\rho = 30$ {m³}; with a waterplane area $LW = 100$ {m²} the draft is 0.30 {m}.

(b) The extra 1.5×10^4 {kg} adds 15 {m³} of displacement, i.e. 0.15 {m} more draft (total 0.45 {m}) if the load is centred. Placed at one end, the barge also acquires a trim angle: the centre of buoyancy must move under the combined centre of mass, so the end carrying the crane sits deeper. With a 20 {m} length and a load at 10 {m} from the centre, the trim is of order $(10\times0.15/20)\approx 4^\circ$ — noticeable, but nowhere near swamping for a barge whose freeboard is more than a metre.

(c) While the container hangs, its weight is carried by the cable, so the barge's draft is unchanged. As it is lowered and touches the water, the water supplies an upthrust that grows with the submerged volume, so the cable tension falls; the barge's draft is still unchanged because nothing rests on it. Once the container is set down on the deck, its full weight is transferred to the barge, the tension drops to zero, and the draft rises by $50/100 = 0.50$ {m}, giving a total draft of 0.95 {m}. **Check:** the draft changes only at the moment of load transfer, which is exactly when the barge's weight changes.